

Pr. 4.24

MX

a) ~~$A = uv^T$~~ $\text{row}(A) = \text{span}(\tau_0)$

$\text{rank}(A) = 1$ ~~$A = uv^T$~~

$\tau_i(A) = c_i \tau_0$
 $\tau_0 \neq 0$

$A = \begin{bmatrix} \tau_1 \\ \vdots \\ \tau_n \end{bmatrix}$

we can take a non zero row vector τ_i from

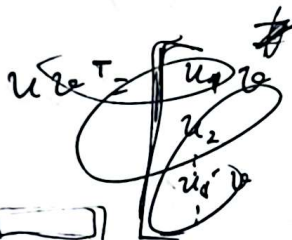
$A = \begin{bmatrix} c_1 \tau_0 \\ \vdots \\ c_n \tau_0 \end{bmatrix} = C \tau^T$ ~~$A = uv^T$~~
we can find such A vector, because other wise $A=0$, $\text{rank}(A)=0$

$C = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$

$\tau = \begin{bmatrix} \tau_{01} \\ \tau_{02} \\ \vdots \\ \tau_{0n} \end{bmatrix}$

$uv^T = \begin{bmatrix} u_1 v^T \\ \vdots \\ u_n v^T \end{bmatrix}$

$\tau^T = [\tau_{01} \dots \tau_{0n}] =$



b)

~~$uv^T x = \lambda x$~~

~~$uv^T(u^T v) = \lambda$~~

$A \neq 0, u \neq 0, v \neq 0$. As $\text{rank}(A) = 1$
 $uv^T x = (u^T v) x$

~~$uv^T u = \lambda u^T u$~~
 $(v^T x) u = (u^T v) x$

if $x = u$

$(v^T u) u = (u^T v) u = (v^T u) u$

c)

$uv^T x = \lambda x$

~~$A = uv^T$~~ has

$\lambda = 0, v^T u$

$(v^T x) u = \lambda x$ $x \neq 0, u \neq 0, v \neq 0$
scalar vector scalar vector

for this to be true, either $x = \kappa u$ or both sides is zero, so, that $\lambda = 0, v^T x = 0$

when $x = \kappa u$, $(\kappa v^T u) u = \lambda \kappa u$
so, $\lambda = v^T u$

because $u \neq 0$, $u^T v$ is eigenvalues

HV.

Pr. 4.24.

d) $\lambda_1 = v^T u$ biggest eigenvalue is $\lambda_1 = v^T u \rightarrow v_1 = u$ ^{eigen vector corresponding to λ_1}
 $\lambda_2 = 0$

we choose random x_0

~~$Ax_0 = uv^T$~~ scalar

$Ax_0 = uv^T x_0 = (v^T x_0)u \rightarrow$ proportional to u ,
 so, 1 iteration
 is enough.

Pr. 4.25.

$$\det(I + uv^T) = 1 + u^T v$$

$$\det(I + uv^T) = \prod_{i=1}^n \lambda_i$$

$$(I + uv^T)x = \lambda x \rightarrow \text{for this to be true,}$$

$$uv^T x \neq x = \lambda x$$

$$uv^T x = (\lambda - 1)x$$

$\lambda - 1$ should be either 0, or $u^T v$

$$\lambda = 1, u^T v + 1$$

$$\det(I + uv^T) = 1 \cdot (u^T v + 1) = u^T v + 1$$

Pr. 4.28.

$$\rho(A) < 1$$

$$(I - A)x = 0$$

we need to prove that x can only be 0.

let's assume contradiction

$$(I - A)x = 0 \quad x \neq 0 \quad x - Ax = 0$$

$Ax = x$ one eigenvalue is 1.

so, $\lambda = 1$, where $x \neq 0$

as $\rho(A) < 1$, it is a contradiction.

$$(I - A)^{-1} = \sum_{k=0}^{\infty} A^k = I + A + A^2 + \dots$$

$$(I - A)\left(\sum_{k=0}^{\infty} A^k\right) = \sum_{k=0}^{\infty} A^k - \sum_{k=0}^{\infty} A^{k+1} =$$

$$= \lim_{n \rightarrow \infty} \left(\sum_{k=0}^n A^k - \sum_{k=0}^n A^{k+1} \right) = \lim_{n \rightarrow \infty} \left(\sum_{k=0}^n A^k - \sum_{k=1}^{n+1} A^k \right) =$$

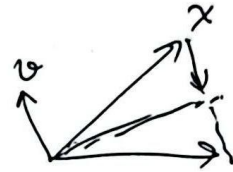
Pr. 4.32.

$$a) H = I - 2 \frac{vv^T}{v^T v}$$

$$Hx = \lambda x$$

$$Hv = \lambda v$$

$$\left(I - 2 \frac{vv^T}{v^T v} \right) x = \lambda x$$



$$x - 2 \frac{vv^T x}{v^T v} = \lambda x$$

$$\|Hx\| = \|x\|$$

$$\|Hx\| = \|\lambda x\| = \|\lambda\| \|x\| = \|\lambda\| \|x\| = \|x\|$$

$$|\lambda| = 1$$

$$(1 - \lambda)x = \frac{2vv^T x}{v^T v}$$

$$b) S = \begin{bmatrix} c & s \\ -s & c \end{bmatrix}$$

$$2vv^T x = (v^T v)(1 - \lambda)x$$

$$x = \frac{2vv^T}{2vv^T} x$$

$$Sx = \lambda x$$

$$(vv^T) x = \frac{v^T v}{2} (1 - \lambda)x$$

from problem 4.24 c)

we know that, $\frac{v^T v}{2} (1 - \lambda) =$

$$\frac{v^T v (1 - \lambda)}{2} = 0 \rightarrow \lambda = 1$$

$$\{0, v^T v\}$$

we chose

$$v = u,$$

VLOS,

A is still

rank one.

$$\frac{v^T v (1 - \lambda)}{2} = v^T v.$$

$$1 - \lambda = 2$$

$$\lambda = -1$$

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$$Gx = \lambda x \quad x \neq 0$$

$$(G - \lambda I)x = 0$$

$$\lambda = c + is, c - is.$$

$$\begin{bmatrix} c - \lambda & s \\ -s & c - \lambda \end{bmatrix} x = 0$$

$$\det \begin{vmatrix} c - \lambda & s \\ -s & c - \lambda \end{vmatrix} = 0$$

$$(c - \lambda)^2 + s^2 = 0$$

$$(c - \lambda)^2 = -s^2$$

$$c - \lambda = \pm is$$

$$\cancel{\lambda = c + is} \quad \lambda = c \mp is.$$

b) $(I-A) \left(\sum_{k=0}^{\infty} A^k \right) = \lim_{n \rightarrow \infty} (I - A^{n+1})$

Assumption:

$\lim_{n \rightarrow \infty} A^{n+1} = 0.$

or

proof.

$\rho(A) < 1$

$\lim_{n \rightarrow \infty} \rho(A)^{n+1} = \lim_{n \rightarrow \infty} \rho$

~~$\lim_{n \rightarrow \infty} \rho(A^{n+1}) = \rho(A)$~~

$\lim_{n \rightarrow \infty} \left[\rho(A^{n+1}) \right] = \lim_{n \rightarrow \infty} \left[(\rho(A))^{n+1} \right] = 0$

Because $\rho(A)$ is

biggest eigen value, as

$n \rightarrow \infty$, all eigen values should go to zero.

so, det

solutions of $\lim_{n \rightarrow \infty} \det(A^{n+1} - \lambda I) = 0$

are all $\lambda = 0$, when $n \rightarrow \infty$

~~$\lim_{n \rightarrow \infty} \lambda^{n+1}$~~

as $n \rightarrow \infty$

$\det(A^{n+1} - \lambda I) = 0$

so, $\rho(A) = \lambda^n = 0$

because of Cayley-Hamilton theorem

$\rho(A) = 0 = A^n$

$\lim_{n \rightarrow \infty} A^{n+1} = 0$

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$$Gx = \lambda x \quad x \neq 0$$

$$(G - \lambda I)x = 0$$

$$\lambda = c + is, c - is.$$

$$\begin{bmatrix} c-\lambda & s \\ -s & c-\lambda \end{bmatrix} x = 0$$

$$\det \begin{vmatrix} c-\lambda & s \\ -s & c-\lambda \end{vmatrix} = 0$$

$$(c-\lambda)^2 + s^2 = 0$$

$$(c-\lambda)^2 = -s^2$$

$$c-\lambda = \pm is$$

$$\cancel{\lambda = c + is} \quad \lambda = c \mp is.$$

Pr. 4.27. . AP

$$6) (I-A) \left(\sum_{k=0}^{\infty} A^k \right) = \lim_{n \rightarrow \infty} (I - A^{n+1})$$

Claim:

$$\text{if } \rho(A) < 1. \rightarrow \lim_{n \rightarrow \infty} A^{n+1} = 0$$

From Jordan normal form theorem, $\exists V, J \in \mathbb{C}^{n \times n}$.where $\det V \neq 0$ ~~det~~.

so, that.

$$A = VJV^{-1}$$

$$J = \begin{bmatrix} J_{m_1}(\lambda_1) & & & \\ & J_{m_2}(\lambda_2) & & \\ & & \ddots & \\ & & & J_{m_{s-1}}(\lambda_{s-1}) \\ & & & & J_{m_s}(\lambda_s) \end{bmatrix}$$

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b) second sol. for $\lim_{n \rightarrow \infty} A^{n+1} = 0$

$$J_{m_i}(\lambda_i) = \begin{bmatrix} \lambda_i & 1 & 0 & \dots & 0 \\ 0 & \lambda_i & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_i & 1 \\ 0 & 0 & 0 & \dots & 0 & \lambda_i \end{bmatrix} \in \mathbb{C}^{m_i \times m_i}, \quad 1 \leq i \leq s.$$

$$A^K = V J^K V^{-1}.$$

$$\text{and } J^K = \begin{bmatrix} J_{m_1}(\lambda_1)^K & & \\ & J_{m_2}(\lambda_2)^K & \\ & & \ddots \\ & & & J_{m_s}(\lambda_s)^K \end{bmatrix}$$

$$J_{m_i}(\lambda_i) = \lambda_i I + M$$

$$J_{m_i}^K(\lambda_i) = (\lambda_i I + M)^K, \quad M = \begin{bmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ & & & 0 \end{bmatrix}$$

Let's use induction.
to prove.

$$M^K =$$

$$M = \begin{bmatrix} 0 & u \\ 0 & 0 \end{bmatrix}$$

$$M^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$M^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & u \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$M = \begin{bmatrix} 0 & u \\ 0 & 0 \end{bmatrix}$$

$$M = \begin{bmatrix} 0 & u \\ 0 & 0 \end{bmatrix}$$

$$M^{k+1} = M^k M.$$

we are using induction to prove.

$$M(\epsilon, i+k) = 1.$$

$$(M^{k+1})_{ij} = \sum_{\ell} c_{\ell}(M^k) c_{\ell}(M) = \begin{cases} 1, & \text{when } i+k = j-1 \\ 0, & \text{other where.} \end{cases}$$

entry of
only $i+k$ $c_{\ell}(M^k) = 1.$

and, $j-1$ entry of $c_{\ell}(M) = 0$

so, $i+k+1 = j,$

$(M^{k+1})_{ij} = 1$, or else,

$$\sum_{r=0}^k \binom{k}{r} \lambda_i^{k-r} M^r =$$

using this,

$(M^{k+1})_{ij} = 0.$

~~$(i, i+a)$~~ $\sum_{r=0}^k \binom{k}{r} \lambda_i^{k-r} M^r = \binom{k}{a} \lambda_i^{k-a}$

when $k \rightarrow \infty$

when $a < 0$, $\sum_{r=0}^k \binom{k}{r} \lambda_i^{k-r} M^r = 0.$

$\lim_{k \rightarrow \infty} \sum_{r=0}^k \binom{k}{r} \lambda_i^{k-r} M^r = \binom{k}{a} \lambda_i^{k-a}$ so,

$\sum_{r=0}^k \binom{k}{r} \lambda_i^{k-r} M^r =$

$\lim_{k \rightarrow \infty} \binom{k}{a} \lambda_i^{k-a} = 0.$

so, as $k \rightarrow \infty$ $\sum_{r=0}^k \binom{k}{r} \lambda_i^{k-r} M^r = 0 \rightarrow$ this means,

$\lim_{k \rightarrow \infty} J^k = 0$ so, $\lim_{k \rightarrow \infty} A^k = \lim_{k \rightarrow \infty} V J^k V^{-1} = V \lim_{k \rightarrow \infty} J^k V^{-1} = 0.$